part-1

1. install mininet tool

2. install ryu controller python packages (or) any SDN controller docker image

3. create a custom topology containing 8 switches and any no. of host machines you want, in mininet u can create custom topologies using python scripts (ask chatgpt). Create topology in such a way that, some switches form cycles in a graph and for different links between switches give different cost parameters

4. run the RYU sdn controller (initially a simple controller doesnt do much, later we'll add features to this)

5. in the custom topology script make it like every device connect to our SDN controller

part-2 - TOPOLOGY discovery by SDN

the SDN recieves connections from our switches. using openflow messages u can query them devices for more info. ask about its links. so u'll have list of switches, links.

use any graph drawing package in python, pass the list of vertices and edges to it and make it draw the network as a graph

part-3 - MANIPULATE LINKS

if possible, by sending openflow messages disable some links. re-draw the network graph again

part-4 - DIJKSTRA

now u know the topology, switches, links and the cost of each links. randomly select one switch as root switch. apply dijkstra algorithm. find the shortest cost path for every switch from the root switch and print